GATE DA 2027

Machine Learning

Topic-wise Question Bank

Weighted by what GATE DA 2024–2026 actually asked

163 practice questions + **8** real previous-year questions • **10** topics

MCQ • MSQ • NAT — the three GATE question types

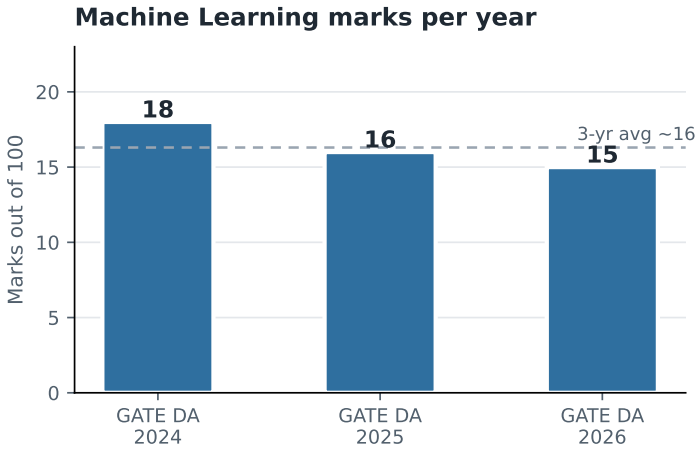
Answer key at the end — answers only, no worked solutions

Every numerical answer in this book was recomputed independently in Python (NumPy) before the key was fixed — all 46 machine checks agreed. The 8 previous-year questions carry the official GATE answer key. Practice options are shuffled, so the key is not all "A".

Prepared for Ashutosh • August 2026

How this book is built

Machine Learning is the largest technical block in GATE DA. Across the official 2024–2026 papers, ML (supervised + unsupervised learning together) has consistently carried the *most* marks of any single subject — roughly **15 to 18 marks** each year. Along with PDSA it decides your rank. The exact figure wobbles because some questions straddle ML and Probability (a Bayes question) or ML and Linear Algebra (PCA, a Gram matrix), so treat the per-year numbers below as ±2.

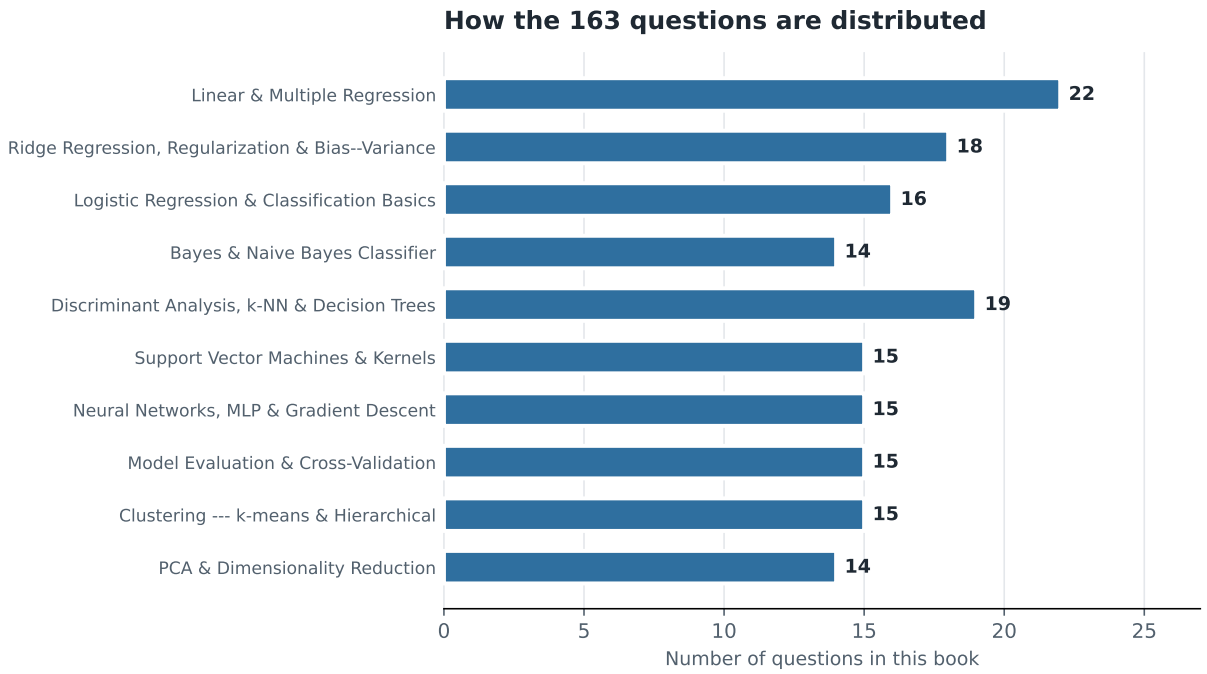


What GATE DA actually asks in ML

Reading every ML question in the official 2024, 2025 and 2026 papers, four habits stand out:

1. **Compute a small quantity by hand.** A precision from a confusion matrix (2026 Q.47), the number of parameters in an MLP (2026 Q.56), a regularized loss (2026 Q.55), the first merge in hierarchical clustering (2026 Q.36). Bring a clear head for arithmetic, not proofs.
2. **Know the *property*, not the derivation.** "Which statement about Ridge Regression is true?" (2026 Q.37); "match PCA / Naive Bayes / Logistic Regression to dimensionality-reduction / generative / discriminative" (2024 Q.18). One clean fact per model.
3. **Reason about geometry.** PCA components are orthogonal so the angle between two of them is 90° (2026 Q.11); a k-means (Voronoi) cell is convex, so a midpoint of two same-cluster points is in the same cluster (2024 Q.19).
4. **Overlap with Probability and Linear Algebra is constant.** Bayes/Naive Bayes is shared with Probability; PCA, SVD and quadratic forms are shared with Linear Algebra. Time spent here is paid two or three times over.

The number of questions per topic below follows that reality: regression, evaluation, clustering, PCA and neural networks — the areas GATE DA returns to every year — get the most space.

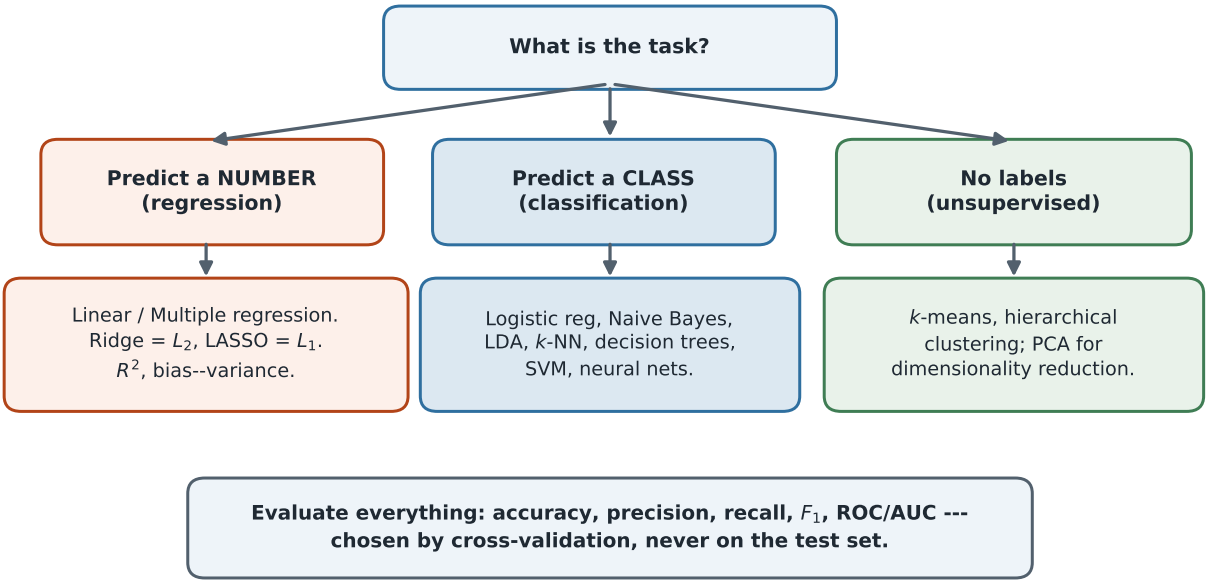


Question types and marking (same as the real paper)

Type	Description	Negative marking
MCQ	four options, exactly one correct	$-1/3$ (1-mark) or $-2/3$ (2-mark)
MSQ	four options, one *or more* correct; no partial credit	none
NAT	numerical answer, entered on a virtual keypad	none

This bank is 65% MCQ, 7% MSQ and 28% NAT. The extra NAT weighting is deliberate: NAT carries no negative marking, and "compute this precision / loss / parameter count" is exactly the drill GATE DA rewards.

One decision, five families of models



Sixteen facts that answer a disproportionate number of ML questions.

- OLS line passes through $(\bar{x}, \bar{y})$; $R^2 = 1 - \text{RSS/TSS}$, and in simple regression $R^2 = r^2$.
- Ridge = L_2 penalty (shrinks, keeps all features); LASSO = L_1 (sparse, selects features).
- Bias–variance: more λ / less flexibility $\Rightarrow$ more bias, less variance.
- Logistic regression: $P(y{=}1) = \sigma(w^{\mathsf{T}}x + b)$, linear boundary, trained by cross-entropy; $\sigma(0) = 0.5$, $\sigma' = \sigma(1-\sigma)$.
- Naive Bayes: features conditionally independent given the class; generative; MAP decision.
- Entropy of a 50/50 node = 1 bit; Gini of 50/50 = 0.5; pure node = 0.
- Information gain = parent entropy − weighted child entropies.
- k-NN: lazy, needs feature scaling; small k $\Rightarrow$ high variance.
- LDA: shared covariance $\Rightarrow$ linear boundary; QDA $\Rightarrow$ quadratic.
- SVM: maximizes margin = $2/\|w\|$; only support vectors matter; kernel trick = inner products.
- MLP parameter count: sum of (units in $\times$ units out) per layer, plus one bias per neuron.
- A stack of purely linear layers collapses to one linear model; XOR needs a hidden layer.
- Accuracy = $(TP{+}TN)$/all; precision = $TP/(TP{+}FP)$; recall = $TP/(TP{+}FN)$; F_1 is their harmonic mean.
- Choose hyperparameters by cross-validation; never touch the test set while tuning.
- k-means minimizes within-cluster SSD, converges to a *local* optimum, assumes spherical clusters; cells are convex.
- PCA: eigenvectors of the covariance, ordered by eigenvalue = variance explained; components are orthogonal (90° apart); unsupervised. LDA is the supervised analogue.

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Part A — Practice questions by topic

1. Linear & Multiple Regression

22 questions • 12 MCQ, 2 MSQ, 8 NAT • Suggested time: 40 minutes

Q.1 [MCQ • 1 Mark • Easy]

In simple linear regression $y = \beta_0 + \beta_1 x + \varepsilon$ fitted by ordinary least squares (OLS), the fitted line always passes through which point?

(A) $(0, \bar{y})$ (B) $(\bar{x}, 0)$ (C) $(\bar{x}, \bar{y})$ (D) $(0, 0)$

Q.2 [NAT • 2 Marks • Moderate]

For OLS simple linear regression, the slope estimate is $\hat{\beta}_1 = \dfrac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$. For the data $x = (1, 2, 3, 4, 5)$, $y = (2, 4, 5, 4, 5)$, the value of $\hat{\beta}_1$ (rounded to two decimals) is _____.

Q.3 [NAT • 2 Marks • Moderate]

For the same data $x = (1, 2, 3, 4, 5)$, $y = (2, 4, 5, 4, 5)$, the OLS intercept $\hat{\beta}_0$ (two decimals) is _____.

Q.4 [MCQ • 1 Mark • Easy]

OLS minimizes which quantity?

(A) $\sum_i |y_i - \hat{y}_i|$ (B) $\sum_i (y_i - \hat{y}_i)^2$

(C) $\sum_i (y_i - \hat{y}_i)$ (D) $\max_i |y_i - \hat{y}_i|$

Q.5 [MCQ • 1 Mark • Easy]

The coefficient of determination R^2 for a regression model is 0.81. Which statement is correct?

(A) 81% of the predictions are exactly correct (B) 81% of the variance in y is explained by the model

(C) The model has 81% accuracy (D) The correlation between x and y is 0.81

Q.6 [NAT • 1 Mark • Easy]

In simple linear regression, R^2 equals the square of the Pearson correlation coefficient r between x and y. If $r = -0.9$, then R^2 = _____.

Q.7 [MSQ • 2 Marks • Moderate]

Which of the following are assumptions of the classical linear regression model?

(A) The errors are uncorrelated with each other (B) The errors have constant variance (homoscedasticity)

(C) The errors ε_i have zero mean (D) The predictor x must be normally distributed

Q.8 [MCQ • 1 Mark • Moderate]

A multiple linear regression model with p predictors plus an intercept is fitted to n observations. The design matrix X (including the intercept column) has dimensions

(A) $p \times n$ (B) $(p+1) \times n$ (C) $n \times (p+1)$ (D) $n \times p$

Q.9 [MCQ • 1 Mark • Moderate]

The normal equations for multiple linear regression give the OLS estimate $\hat{\boldsymbol{\beta}} = (X^{\mathsf{T}}X)^{-1}X^{\mathsf{T}}y$. A necessary condition for this closed form to exist is that

(A) y has zero mean

(B) $X^{\mathsf{T}}X$ is invertible (columns of X are linearly independent)

(C) $n = p$

(D) X is a square matrix

Q.10 [MCQ • 1 Mark • Moderate]

If a predictor is perfectly collinear with another predictor in a multiple regression, then

(A) the model automatically drops one predictor

(B) $X^{\mathsf{T}}X$ is singular and the OLS estimate is not unique

(C) the residuals become zero

(D) R^2 becomes exactly 1

Q.11 [MCQ • 1 Mark • Moderate]

Adding an extra predictor to a multiple linear regression model, keeping the same data, can never ___ the training R^2.

(A) change (B) equal (C) decrease (D) increase

Q.12 [MCQ • 1 Mark • Easy]

Which quantity, unlike ordinary R^2, can decrease when a useless predictor is added?

(A) The intercept (B) Number of observations

(C) Total sum of squares (D) Adjusted R^2

Q.13 [NAT • 1 Mark • Easy]

For a fitted regression, the residuals are $e = (1, -2, 0, 1, -1, 1)$. The residual sum of squares (RSS) is _____.

Q.14 [NAT • 1 Mark • Easy]

Consider data with total sum of squares TSS = 200 and residual sum of squares RSS = 50. The R^2 of the model is _____.

Q.15 [MCQ • 2 Marks • Moderate]

In gradient descent for linear regression with squared-error loss $J(\beta) = \frac{1}{2n}\sum(y_i - x_i^\mathsf{T}\beta)^2$, the gradient with respect to β is

(A) $-X^\mathsf{T}y$

(B) $-\frac{1}{n}X^\mathsf{T}(y - X\beta)$

(C) $\frac{1}{n}X^\mathsf{T}X\beta$

(D) $-\frac{1}{n}X(y - X\beta)$

Q.16 [NAT • 1 Mark • Easy]

A regression is fit as $\hat{y} = 3 + 2x_1 - x_2$. For $x_1 = 4$, $x_2 = 5$, the predicted value $\hat{y}$ is _____.

Q.17 [MSQ • 2 Marks • Moderate]

Which statement about the OLS residuals $e_i = y_i - \hat{y}_i$ is always true when the model includes an intercept?

(A) $\sum_i e_i = 0$

(B) All residuals are non-negative

(C) The residuals are orthogonal to each column of X

(D) $\sum_i \hat{y}_i = \sum_i y_i$

Q.18 [MCQ • 1 Mark • Moderate]

Standardizing the predictors (mean 0, variance 1) before OLS changes which of the following?

(A) The magnitude of the coefficients, but not the fitted values $\hat{y}$

(B) The R^2

(C) Nothing at all

(D) The fitted values $\hat{y}$

Q.19 [NAT • 1 Mark • Easy]

A dataset has $x = (0, 1, 2, 3)$ and $y = (1, 3, 5, 7)$, which lie exactly on a line. The RSS of the OLS fit is _____.

Q.20 [NAT • 1 Mark • Easy]

For $x = (0, 1, 2, 3)$, $y = (1, 3, 5, 7)$, the OLS slope $\hat{\beta}_1$ is _____.

Q.21 [MCQ • 1 Mark • Moderate]

Heteroscedasticity in a linear regression means that

(A) the predictors are correlated

(B) the error variance is not constant across observations

(C) the model is nonlinear

(D) the errors have non-zero mean

Q.22 [MCQ • 1 Mark • Moderate]

You fit $y = \beta_0 + \beta_1 x + \beta_2 x^2$ by OLS. This model is

(A) linear in the parameters and can be solved by ordinary least squares

(B) not a regression model

(C) solvable only with regularization

(D) nonlinear in the parameters and needs gradient descent

2. Ridge Regression, Regularization & Bias–Variance

18 questions • 17 MCQ, 1 MSQ, 0 NAT • Suggested time: 33 minutes

Q.1 [MCQ • 1 Mark • Easy]

Ridge regression minimizes $\sum_i (y_i - x_i^\mathsf{T}\beta)^2 + \lambda\|\beta\|_2^2$. The penalty term uses which norm?

(A) L_2 norm (B) L_0 norm (C) L_1 norm (D) L_∞ norm

Q.2 [MCQ • 2 Marks • Moderate]

Which statement about Ridge Regression is true? *(Based on GATE DA 2026, Q.37.)*

(A) The regularizer guards against large parameter values / over-fitting

(B) The regularizer of Ridge Regression uses the L_1 norm

(C) Ridge Regression always reduces the bias of the model

(D) Ridge Regression reduces the number of parameters to exactly zero

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Q.3 [MCQ • 1 Mark • Moderate]

As the ridge penalty $\lambda \to \infty$, the ridge coefficient estimates $\hat{\beta}$ (excluding the intercept) tend to

(A) 1 (B) ∞ (C) the OLS estimate (D) 0

Q.4 [MCQ • 1 Mark • Easy]

At $\lambda = 0$, ridge regression coincides with

(A) LASSO (B) the null model

(C) a constant predictor (D) ordinary least squares

Q.5 [MCQ • 1 Mark • Moderate]

Which regularizer tends to produce *sparse* solutions (many coefficients exactly zero), effectively performing feature selection?

(A) no penalty (B) Ridge (L_2) (C) L_2^2 (D) LASSO (L_1)

Q.6 [MCQ • 2 Marks • Moderate]

The ridge estimate is $\hat{\beta}_{\text{ridge}} = (X^\mathsf{T}X + \lambda I)^{-1}X^\mathsf{T}y$. Adding λI helps mainly because it

(A) makes the estimate unbiased

(B) makes $X^\mathsf{T}X + \lambda I$ invertible even when $X^\mathsf{T}X$ is singular

(C) guarantees zero training error

(D) removes the need for an intercept

Q.7 [MCQ • 1 Mark • Moderate]

Increasing the regularization strength λ in ridge regression generally

(A) decreases both bias and variance
(B) increases both bias and variance
(C) increases bias and decreases variance
(D) decreases bias and increases variance

Q.8 [MCQ • 1 Mark • Moderate]

The expected test error of a model can be decomposed as

(A) $\text{Bias}^2 - \text{Variance}$
(B) $\text{Bias}^2 + \text{Variance} + \text{Irreducible error}$
(C) $\text{Variance} - \text{Bias}^2$
(D) $\text{Bias} + \text{Variance}$

Q.9 [MCQ • 1 Mark • Easy]

A very flexible model that fits the training data almost perfectly but generalizes poorly is said to have

(A) high bias and high variance
(B) high variance and low bias
(C) high bias and low variance
(D) low bias and low variance

Q.10 [MSQ • 2 Marks • Moderate]

Which of the following typically *increase* model variance?

(A) Increasing the regularization strength
(B) Increasing model complexity / flexibility
(C) Reducing the regularization strength
(D) Decreasing the amount of training data

Q.11 [MCQ • 1 Mark • Easy]

Underfitting corresponds to

(A) high bias (B) low bias (C) zero training error (D) high variance

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Q.12 [MCQ • 2 Marks • Moderate]

Consider the linear ridge objective at a point where $w = [w_1, w_2]$. The L_2 regularizer contributes a gradient term proportional to

(A) $2\lambda w$
(B) $\lambda \,\text{sign}(w)$
(C) 0
(D) λ

Q.13 [MCQ • 1 Mark • Moderate]

For the elastic net penalty $\lambda_1\|\beta\|_1 + \lambda_2\|\beta\|_2^2$, setting $\lambda_2 = 0$ recovers

(A) LASSO (B) OLS (C) the null model (D) Ridge

Q.14 [MCQ • 1 Mark • Moderate]

As training-set size $n \to \infty$ (model fixed), the variance term of a learning algorithm generally

(A) increases without bound
(B) stays exactly constant

(C) equals the bias (D) decreases toward zero

Q.15 [MCQ • 1 Mark • Easy]

Which is the best practical tool to *choose* the ridge penalty λ?

(A) k-fold cross-validation (B) maximizing training R^2

(C) setting $\lambda = 1$ always (D) minimizing the number of predictors

Q.16 [MCQ • 1 Mark • Easy]

A model with training error 2% and test error 30% most likely suffers from

(A) underfitting (high bias) (B) data leakage into the test set

(C) high irreducible error (D) overfitting (high variance)

Q.17 [MCQ • 1 Mark • Moderate]

Ridge regression shrinks coefficients but, unlike LASSO, generally does *not*

(A) set coefficients exactly to zero (B) increase bias

(C) reduce variance (D) depend on λ

Q.18 [MCQ • 1 Mark • Moderate]

Early stopping in iterative training acts as a form of

(A) feature scaling (B) regularization (C) data augmentation (D) loss function

3. Logistic Regression & Classification Basics

16 questions • 12 MCQ, 1 MSQ, 3 NAT • Suggested time: 30 minutes

Q.1 [MCQ • 1 Mark • Easy]

Logistic regression models $P(y = 1 \mid x)$ using which function of the linear score $z = w^{\mathsf{T}}x + b$?

(A) $\max(0, z)$ (ReLU)
(B) identity z
(C) z^2
(D) sigmoid $\sigma(z) = \frac{1}{1+e^{-z}}$

Q.2 [NAT • 1 Mark • Easy]

The sigmoid $\sigma(z) = \frac{1}{1+e^{-z}}$ evaluated at $z = 0$ equals _____.

Q.3 [MCQ • 1 Mark • Moderate]

Logistic regression is trained by minimizing which loss?

(A) mean squared error
(B) binary cross-entropy (log loss)
(C) 0–1 loss
(D) hinge loss

Q.4 [MCQ • 1 Mark • Moderate]

In logistic regression, the decision boundary $P(y = 1 \mid x) = 0.5$ corresponds to

(A) a quadratic boundary
(B) $w^{\mathsf{T}}x + b = 1$
(C) $w^{\mathsf{T}}x + b = 0$, a linear boundary
(D) $x = 0$

Q.5 [MCQ • 1 Mark • Moderate]

The log-odds (logit) in logistic regression, $\log \frac{P(y=1|x)}{P(y=0|x)}$, equals

(A) $w^{\mathsf{T}}x + b$
(B) $\frac{1}{2}$
(C) $e^{w^{\mathsf{T}}x+b}$
(D) $\sigma(w^{\mathsf{T}}x + b)$

Q.6 [MCQ • 1 Mark • Easy]

If the fitted logistic model gives score $z = w^{\mathsf{T}}x + b = 2$, the predicted probability $\sigma(2)$ is closest to

(A) 0.98 (B) 0.12 (C) 0.88 (D) 0.5

Q.7 [NAT • 2 Marks • Moderate]

Cross-entropy loss for a single example with true label $y = 1$ and predicted probability $\hat{p}$ is $-\log \hat{p}$. If $\hat{p} = 0.25$, the loss (natural log, two decimals) is _____.

Q.8 [MSQ • 2 Marks • Moderate]

Which statements about logistic regression are correct?

(A) It is a linear classifier in feature space

(B) Its output can be interpreted as a probability

(C) It is trained by maximum likelihood

(D) It minimizes squared error between labels and predictions

Q.9 [MCQ • 1 Mark • Easy]

For a balanced two-class problem, using a probability threshold of 0.5 on a logistic model, an example is classified as class 1 when

(A) $w^{\mathsf{T}}x + b < 0$

(B) $\sigma(z) < 0.5$

(C) $w^{\mathsf{T}}x + b > 0$

(D) $z = 0$

Q.10 [MCQ • 1 Mark • Moderate]

Softmax regression generalizes logistic regression to K classes. The predicted probabilities across the K classes always

(A) are all equal (B) sum to 0 (C) sum to K (D) sum to 1

Q.11 [MCQ • 1 Mark • Moderate]

The derivative of the sigmoid satisfies $\sigma'(z) =$

(A) $\sigma(z)$

(B) $1 - \sigma(z)$

(C) $\sigma(z)\,(1 - \sigma(z))$

(D) $\sigma(z)^2$

Q.12 [NAT • 1 Mark • Easy]

Odds of 3:1 in favour of an event correspond to a probability of _____.

Q.13 [MCQ • 2 Marks • Moderate]

A logistic regression coefficient $w_j = 0.7$ means that a one-unit increase in feature x_j multiplies the odds of $y = 1$ by

(A) $\log 0.7$

(B) 1.7

(C) $e^{0.7} \approx 2.01$

(D) 0.7

Q.14 [MCQ • 1 Mark • Moderate]

Which is a reason to prefer cross-entropy over squared error for training logistic regression?

(A) Squared error is always larger

(B) The cross-entropy loss is convex in the parameters, giving a well-behaved optimization

(C) Squared error cannot be computed for probabilities

(D) Cross-entropy needs no gradient

Q.15 [MCQ • 1 Mark • Moderate]

A generative classifier models $P(x \mid y)$ and $P(y)$, whereas a discriminative classifier such as logistic regression models

(A) $P(x, y)$ only

(B) nothing probabilistic

(C) $P(y \mid x)$ directly

(D) $P(x)$ only

Q.16 [MCQ • 2 Marks • Hard]

If all features are scaled by a large constant before logistic-regression training with no regularization, the learned decision boundary (as a set in feature space)

(A) is unchanged (weights rescale to compensate)

(B) becomes nonlinear

(C) disappears

(D) rotates

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4. Bayes & Naive Bayes Classifier

14 questions • 7 MCQ, 1 MSQ, 6 NAT • Suggested time: 28 minutes

Q.1 [MCQ • 1 Mark • Easy]

Bayes' theorem states $P(y \mid x) = \dfrac{P(x \mid y)\,P(y)}{P(x)}$. The term $P(y)$ is called the

(A) likelihood

(B) prior probability

(C) evidence

(D) posterior probability

Q.2 [NAT • 2 Marks • Hard]

A disease affects 1% of a population. A test is 90% sensitive (true-positive rate) and has a 5% false-positive rate. Given a positive test, the probability of actually having the disease (two decimals) is _____.

Q.3 [MCQ • 1 Mark • Easy]

The 'naive' assumption in a Naive Bayes classifier is that

(A) features are conditionally independent given the class

(B) the data are normally distributed

(C) features are independent of the class

(D) the classes are equally likely

Q.4 [MCQ • 1 Mark • Moderate]

A Naive Bayes classifier assigns the class $\hat{y} = \arg\max_y P(y) \prod_j P(x_j \mid y)$. This decision rule is called

(A) maximum a posteriori (MAP)

(B) maximum likelihood only

(C) nearest neighbour

(D) least squares

Q.5 [NAT • 2 Marks • Hard]

Two fair coins and one biased coin (heads probability 0.9) are in a bag. A coin is drawn at random and flipped, giving heads. The probability it was the biased coin (two decimals) is _____.

Q.6 [MCQ • 1 Mark • Moderate]

In text classification with Naive Bayes, 'Laplace (add-one) smoothing' is used mainly to

(A) avoid zero probabilities for words unseen in a class during training

(B) reduce the number of features

(C) speed up training

(D) make features independent

Q.7 [NAT • 1 Mark • Easy]

$P(A) = 0.3$, $P(B) = 0.5$, and A, B independent. Then $P(A \cap B) =$ _____.

Q.8 [MCQ • 1 Mark • Moderate]

For a two-class Naive Bayes with equal priors, the classifier reduces to choosing the class with the larger

(A) evidence $P(x)$

(B) likelihood $\prod_j P(x_j \mid y)$

(C) prior $P(y)$

(D) posterior variance

Q.9 [MSQ • 2 Marks • Moderate]

Which are true of Gaussian Naive Bayes?

(A) It estimates a per-class mean and variance for each feature

(B) It requires a linear kernel

(C) Features are assumed conditionally independent given the class

(D) Each feature is modelled as Gaussian within each class

Q.10 [NAT • 2 Marks • Moderate]

Given $P(\text{spam}) = 0.4$, $P(\text{`win'} \mid \text{spam}) = 0.5$, $P(\text{`win'} \mid \text{ham}) = 0.1$, the posterior $P(\text{spam} \mid \text{`win'})$ (two decimals) is _____.

Q.11 [NAT • 1 Mark • Easy]

If two events are mutually exclusive with $P(A) = 0.2$, $P(B) = 0.3$, then $P(A \cup B) =$ _____.

Q.12 [MCQ • 2 Marks • Hard]

The Naive Bayes classifier's decision boundary between two Gaussian classes with equal covariance $\sigma^2 I$ is

(A) quadratic (B) linear (C) nonexistent (D) circular

Q.13 [NAT • 1 Mark • Moderate]

A bag has 3 red and 2 blue balls. Two balls are drawn without replacement. The probability both are red (two decimals) is _____.

Q.14 [MCQ • 1 Mark • Moderate]

Compared with a full joint model over d binary features, Naive Bayes needs far fewer parameters because it stores

(A) $O(2^d)$ parameters

(B) only the prior

(C) no parameters

(D) $O(d)$ per-feature conditionals instead of $O(2^d)$ joint entries

5. Discriminant Analysis, k-NN & Decision Trees

19 questions • 9 MCQ, 1 MSQ, 9 NAT • Suggested time: 36 minutes

Q.1 [NAT • 1 Mark • Easy]

The entropy of a node with class proportions $(\frac{1}{2}, \frac{1}{2})$, using $\log_2$, is _____ bit(s).

Q.2 [NAT • 1 Mark • Easy]

The Gini impurity of a node with class proportions $(0.5, 0.5)$ is _____.

Q.3 [NAT • 1 Mark • Moderate]

A node has 8 positive and 2 negative examples. Its Gini impurity (two decimals) is _____.

Q.4 [NAT • 2 Marks • Moderate]

The entropy of a node with 8 positive and 2 negative examples ($\log_2$, two decimals) is _____.

Q.5 [MCQ • 1 Mark • Moderate]

Information gain of a split is defined as

(A) the product of child entropies

(B) parent entropy minus the weighted average of child entropies

(C) child entropy minus parent entropy

(D) the sum of child entropies

Q.6 [NAT • 2 Marks • Moderate]

A parent node (entropy 1.0) with 10 examples is split into a pure node of 6 examples (entropy 0) and a node of 4 examples (entropy 1.0). The information gain (two decimals) is _____.

Q.7 [MCQ • 1 Mark • Moderate]

k-nearest-neighbour (k-NN) classification with $k = 1$ has

(A) zero training error but potentially high variance

(B) high training error

(C) no hyperparameters

(D) a linear decision boundary

Q.8 [MCQ • 1 Mark • Moderate]

In k-NN, increasing k generally

(A) always improves accuracy

(B) smooths the decision boundary and reduces variance

(C) increases variance

(D) reduces bias

Q.9 [MCQ • 2 Marks • Moderate]

Points: class A at $(0, 0)$ and $(1, 1)$; class B at $(3, 3)$ and $(4, 4)$. Using 1-NN with Euclidean distance, the query point $(2, 2)$ is nearest to a point of class

(A) both equally — undefined

(B) A (tie broken by $(1, 1)$)

(C) neither

(D) B

Q.10 [NAT • 1 Mark • Easy]

The Euclidean distance between $(1, 2, 2)$ and $(4, 6, 2)$ is _____.

Q.11 [NAT • 1 Mark • Easy]

The Manhattan (L_1) distance between $(1, 2)$ and $(4, 6)$ is _____.

Q.12 [MCQ • 1 Mark • Easy]

k-NN is called a *lazy* learner because it

(A) trains very slowly

(B) does no work at training time and defers computation to prediction time

(C) uses few features

(D) ignores the labels

Q.13 [MCQ • 1 Mark • Moderate]

Why is feature scaling important for k-NN?

(A) Distances are dominated by features with larger numeric ranges if unscaled

(B) k-NN cannot handle negative numbers

(C) Scaling changes the labels

(D) It reduces the number of neighbours

Q.14 [MCQ • 1 Mark • Moderate]

Linear Discriminant Analysis (LDA) assumes the classes are Gaussian with

(A) different covariance matrices

(B) diagonal covariance only

(C) zero mean

(D) a shared (common) covariance matrix

Q.15 [MCQ • 1 Mark • Moderate]

LDA produces a decision boundary that is ___, whereas Quadratic Discriminant Analysis (QDA) can produce a ___ boundary.

(A) linear; quadratic (B) quadratic; linear (C) linear; circular (D) circular; linear

---

Q.16 [MSQ • 2 Marks • Moderate]

Which statements about decision trees are correct?

(A) They require feature scaling to work

(B) A fully grown tree can overfit the training data

(C) They can model nonlinear decision boundaries

(D) Pruning reduces variance

Q.17 [MCQ • 1 Mark • Moderate]

A decision-tree split on a purely random feature (no relation to the label) has expected information gain approximately

(A) 1 (B) negative (C) maximal (D) 0

Q.18 [NAT • 1 Mark • Easy]

The entropy of a pure node (all examples one class) is _____.

Q.19 [NAT • 2 Marks • Moderate]

For a 3-class node with proportions $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, the entropy in bits (two decimals) is _____.

6. Support Vector Machines & Kernels

15 questions • 10 MCQ, 1 MSQ, 4 NAT • Suggested time: 31 minutes

Q.1 [MCQ • 1 Mark • Easy]

A linear SVM finds the separating hyperplane that

(A) passes through the class means

(B) minimizes the number of support vectors

(C) minimizes the training loss only

(D) maximizes the margin between the classes

Q.2 [MCQ • 1 Mark • Moderate]

The 'support vectors' of a trained SVM are

(A) the test points

(B) the class centroids

(C) the training points lying on (or inside) the margin

(D) all training points

Q.3 [MCQ • 2 Marks • Moderate]

For a hard-margin SVM with hyperplane $w^{\mathsf{T}}x + b = 0$, the margin width equals

(A) $\|w\|$ (B) $2\|w\|$ (C) $\dfrac{2}{\|w\|}$ (D) $\dfrac{1}{\|w\|}$

Q.4 [MCQ • 1 Mark • Moderate]

The hard-margin SVM primal minimizes $\frac{1}{2}\|w\|^2$ subject to $y_i(w^{\mathsf{T}}x_i + b) \geq 1$. Minimizing $\|w\|^2$ is equivalent to

(A) maximizing the margin

(B) maximizing training error

(C) removing support vectors

(D) minimizing the margin

Q.5 [MCQ • 2 Marks • Moderate]

The soft-margin SVM introduces slack variables ξ_i and a parameter C. A very large C

(A) ignores the data

(B) allows many misclassifications

(C) forces a linear kernel

(D) penalizes misclassification heavily, giving a narrower margin (risk of overfitting)

Q.6 [MCQ • 1 Mark • Moderate]

The 'kernel trick' lets an SVM

(A) reduce the number of classes

(B) compute inner products in a high-dimensional feature space without explicit mapping

(C) avoid using any features

(D) eliminate support vectors

Q.7 [MSQ • 2 Marks • Moderate]

Which of the following are valid SVM kernels?

(A) RBF (Gaussian) kernel $\exp(-\gamma\|x - z\|^2)$

(B) Polynomial kernel $(x^\mathsf{T} z + c)^d$

(C) Linear kernel $x^\mathsf{T} z$

(D) $\|x - z\|$ (raw Euclidean distance)

Q.8 [NAT • 1 Mark • Easy]

For two points $x = (1, 2)$ and $z = (2, 0)$, the linear-kernel value $x^\mathsf{T} z$ is _____.

Q.9 [NAT • 1 Mark • Moderate]

For $x = (1, 2)$, $z = (2, 0)$ and $c = 1, d = 2$, the polynomial kernel $(x^\mathsf{T} z + c)^2$ is _____.

Q.10 [NAT • 1 Mark • Easy]

The RBF kernel $\exp(-\gamma\|x - z\|^2)$ evaluated when $x = z$ equals _____.

Q.11 [MCQ • 2 Marks • Hard]

A very small γ in an RBF-kernel SVM makes the kernel

(A) very spiky (each point influences only itself)

(B) smooth / almost linear (each point influences a wide region)

(C) equal to a polynomial kernel

(D) undefined

Q.12 [MCQ • 2 Marks • Moderate]

An SVM's decision function depends on the training data only through

(A) inner products (kernels) between data points

(B) the number of features

(C) the labels only

(D) the raw feature values individually

Q.13 [MCQ • 2 Marks • Moderate]

Removing a training point that is *not* a support vector from an SVM's training set

(A) does not change the learned decision boundary

(B) increases the margin

(C) always changes the boundary

(D) makes the problem infeasible

Q.14 [MCQ • 1 Mark • Moderate]

The hinge loss for an SVM on an example with margin $m = y(w^\mathsf{T} x + b)$ is

(A) m^2 (B) $-\log m$ (C) $\max(0,\, 1 - m)$ (D) $\max(0, m)$

Q.15 [NAT • 1 Mark • Easy]

For a linearly separable dataset, a hard-margin SVM's training error is _____.

7. Neural Networks, MLP & Gradient Descent

15 questions • 10 MCQ, 1 MSQ, 4 NAT • Suggested time: 33 minutes

Q.1 [NAT • 2 Marks • Moderate]

A fully-connected feed-forward network has an input layer of 4 units, one hidden layer of 5 units, and an output layer of 3 units. Counting weights only (no biases), the number of learnable weights is _____.

Q.2 [NAT • 2 Marks • Moderate]

For the same $4 \to 5 \to 3$ network, including one bias per neuron in the hidden and output layers, the total number of learnable parameters is _____.

Q.3 [MCQ • 1 Mark • Easy]

The ReLU activation is defined as

(A) $\frac{1}{1+e^{-z}}$ (B) z^2 (C) $\tanh(z)$ (D) $\max(0, z)$

Q.4 [MCQ • 1 Mark • Moderate]

Backpropagation is fundamentally an efficient application of

(A) the Bayes rule (B) matrix inversion

(C) dynamic programming on graphs (D) the chain rule of differentiation

Q.5 [MCQ • 2 Marks • Moderate]

A multilayer perceptron with only *linear* activation functions in all layers is equivalent to

(A) an SVM with RBF kernel (B) a single linear model

(C) a nonlinear model (D) a decision tree

Q.6 [MCQ • 1 Mark • Moderate]

In stochastic gradient descent (SGD) with learning rate η, the update is $w \leftarrow w - \eta \nabla_w L$. Too large a learning rate typically causes

(A) overfitting only (B) divergence / oscillation of the loss

(C) slow but guaranteed convergence (D) zero gradient

Q.7 [NAT • 2 Marks • Moderate]

One step of gradient descent on $f(w) = w^2$ from $w = 4$ with learning rate $\eta = 0.1$ gives the new $w =$ _____.

Q.8 [MCQ • 1 Mark • Moderate]

The vanishing-gradient problem in deep networks with sigmoid activations refers to

(A) weights becoming exactly zero

(B) gradients blowing up

(C) the loss becoming negative

(D) gradients shrinking toward zero in early layers, stalling learning

---

Q.9 [MSQ • 2 Marks • Moderate]

Which of the following help train deep neural networks?

(A) Careful weight initialization

(B) Using only linear activations

(C) ReLU activations

(D) Batch normalization

Q.10 [MCQ • 1 Mark • Moderate]

A single perceptron (linear threshold unit) cannot represent which Boolean function?

(A) NOT (B) XOR (C) OR (D) AND

Q.11 [MCQ • 1 Mark • Moderate]

Mini-batch gradient descent, compared with full-batch, generally

(A) is always slower

(B) requires the whole dataset per step

(C) cannot use a learning rate

(D) gives noisier but faster (per-update) progress

Q.12 [MCQ • 2 Marks • Hard]

The softmax output layer with cross-entropy loss produces a gradient (w.r.t. the logits) of the simple form

(A) $\log \hat{p}$

(B) $y - \hat{p}^2$

(C) $\hat{p} - y$ (predicted minus one-hot target)

(D) $\hat{p}$

Q.13 [NAT • 1 Mark • Easy]

A network layer computes $a = \sigma(Wx + b)$ with W of shape 5×4. The input x has dimension _____.

Q.14 [MCQ • 1 Mark • Moderate]

Dropout regularizes a neural network by

(A) randomly deactivating units during training

(B) scaling the learning rate

(C) adding an L_1 penalty

(D) removing layers permanently

Q.15 [MCQ • 2 Marks • Hard]

With learning rate η, gradient descent on the convex quadratic $f(w) = \frac{1}{2}w^2$ converges for which range of η?

(A) $\eta < 0$ (B) any η (C) $0 < \eta < 2$ (D) $\eta > 2$

8. Model Evaluation & Cross-Validation

15 questions • 8 MCQ, 1 MSQ, 6 NAT • Suggested time: 28 minutes

Q.1 [NAT • 1 Mark • Easy]

A confusion matrix gives TP= 40, FP= 10, FN= 20, TN= 30. The accuracy (two decimals) is _____.

Q.2 [NAT • 1 Mark • Moderate]

With TP= 40, FP= 10, FN= 20, the precision (two decimals) is _____.

Q.3 [NAT • 1 Mark • Moderate]

With TP= 40, FN= 20, the recall (two decimals) is _____.

Q.4 [MCQ • 1 Mark • Moderate]

The F_1 score is the

(A) harmonic mean of precision and recall (B) product of precision and recall

(C) maximum of precision and recall (D) arithmetic mean of precision and recall

Q.5 [NAT • 2 Marks • Moderate]

If precision $= 0.8$ and recall $= 0.67$, the F_1 score (two decimals) is _____.

Q.6 [MCQ • 1 Mark • Moderate]

For a highly *imbalanced* dataset (e.g. 1% positive), which metric is most misleading?

(A) accuracy (B) F_1 score (C) recall (D) precision

Q.7 [NAT • 1 Mark • Easy]

k-fold cross-validation with $k = 5$ means the data are split into 5 parts and the model is trained _____ times.

Q.8 [MCQ • 1 Mark • Moderate]

Leave-one-out cross-validation (LOOCV) on n examples is k-fold CV with

(A) $k = 1$ (B) $k = n$ (C) $k = \sqrt{n}$ (D) $k = 2$

Q.9 [MCQ • 1 Mark • Moderate]

The ROC curve plots

(A) precision against recall (B) true-positive rate against false-positive rate

(C) accuracy against threshold (D) TP against TN

Q.10 [MCQ • 1 Mark • Easy]

An AUC (area under the ROC curve) of 0.5 indicates a classifier that is

(A) worse than random

(B) perfect

(C) overfitting

(D) no better than random

Q.11 [MSQ • 2 Marks • Moderate]

Which are correct about cross-validation?

(A) It gives a less biased estimate of test error than a single train/test split

(B) Larger k increases computational cost

(C) It uses the test set during training

(D) Every example is used for validation exactly once in k-fold CV

Q.12 [MCQ • 2 Marks • Hard]

Consider a classifier evaluated on a confusion matrix. *(GATE DA 2026, Q.53 pattern.)* For class X: TP$=30$, FP$=10$; for class Y: TP$=15$, FP$=5$. Which statement about precision is correct?

(A) Precision of X (0.75) equals precision of Y (0.75)

(B) Precision cannot be computed

(C) Precision of X is higher than precision of Y

(D) Precision of Y is higher than precision of X

Q.13 [MCQ • 1 Mark • Moderate]

A model is tuned by trying many hyperparameters on the *test* set and picking the best. This is bad practice because it causes

(A) optimistic bias / information leakage from the test set

(B) nothing wrong

(C) high bias

(D) underfitting

Q.14 [MCQ • 1 Mark • Moderate]

Specificity is defined as

(A) $\mathrm{TP}/(\mathrm{TP}+\mathrm{FP})$

(B) $\mathrm{TN}/(\mathrm{TN}+\mathrm{FN})$

(C) $\mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$

(D) $\mathrm{TN}/(\mathrm{TN}+\mathrm{FP})$

Q.15 [NAT • 2 Marks • Moderate]

With TP$=40$, FP$=10$, FN$=20$, TN$=30$, the false-positive rate (two decimals) is _____.

9. Clustering — k-means & Hierarchical

15 questions • 12 MCQ, 1 MSQ, 2 NAT • Suggested time: 27 minutes

Q.1 [MCQ • 1 Mark • Moderate]

The k-means objective minimizes

(A) the maximum cluster diameter

(B) the between-cluster distance

(C) the within-cluster sum of squared distances to the cluster centroids

(D) the number of clusters

Q.2 [MCQ • 1 Mark • Easy]

In Lloyd's k-means algorithm, each iteration performs two steps:

(A) assign points to the nearest centroid, then recompute centroids as cluster means

(B) fit a line, then project points

(C) sort points, then remove outliers

(D) merge the two closest clusters, then split the largest

Q.3 [NAT • 1 Mark • Easy]

Points $2, 4, 6, 10, 12, 14$ are clustered into two groups $\{2, 4, 6\}$ and $\{10, 12, 14\}$. The centroid of the first cluster is _____.

Q.4 [NAT • 1 Mark • Moderate]

For 1-D points $\{2, 4, 6\}$, the within-cluster sum of squared distances to the centroid (mean = 4) is _____.

Q.5 [MCQ • 1 Mark • Moderate]

k-means is guaranteed to

(A) work only in 2-D

(B) produce equal-size clusters

(C) find the global optimum

(D) converge to a local optimum (objective never increases)

Q.6 [MCQ • 1 Mark • Easy]

The result of k-means depends on the initialization. A common remedy is

(A) multiple random restarts / k-means++ initialization

(B) sorting the data

(C) using $k = 1$

(D) removing all features

Q.7 [MCQ • 1 Mark • Moderate]

In agglomerative (bottom-up) hierarchical clustering with *single linkage*, the distance between two clusters is

(A) the distance between centroids

(B) the average of all pairwise distances

(C) the minimum distance between any pair of points across the clusters

(D) the maximum such distance

Q.8 [MCQ • 1 Mark • Moderate]

Complete linkage uses the ___ pairwise distance between clusters, and tends to produce ___ clusters.

(A) maximum; compact (B) minimum; elongated (C) average; spherical (D) minimum; compact

---

Q.9 [MCQ • 2 Marks • Moderate]

Points on a line at positions $1, 2, 5, 7$. Using single-linkage agglomerative clustering, the first two points merged are

(A) 1 and 2 (distance 1) (B) 1 and 7 (C) 2 and 5 (distance 3) (D) 5 and 7 (distance 2)

Q.10 [MSQ • 2 Marks • Moderate]

Which statements about k-means are true?

(A) It is sensitive to feature scaling

(B) It always finds the globally optimal clustering

(C) It requires the number of clusters k to be specified in advance

(D) It can converge to different solutions for different initializations

---

Q.11 [MCQ • 1 Mark • Easy]

Hierarchical clustering produces a ___, which can be cut at a chosen height to obtain a given number of clusters.

(A) ROC curve (B) dendrogram (C) histogram (D) scatter plot

Q.12 [MCQ • 1 Mark • Moderate]

The ‘elbow method’ is used to

(A) scale the features (B) remove outliers

(C) choose the number of clusters k (D) initialize centroids

Q.13 [MCQ • 2 Marks • Moderate]

Points $A(0, 0)$, $B(0, 2)$, $C(5, 0)$. With Manhattan distance and single linkage, the two clusters merged first are

(A) B and C (distance 7) (B) A and C (distance 3)

(C) A and B (distance 2)

(D) A and C (distance 5)

Q.14 [MCQ • 1 Mark • Moderate]

k-means implicitly assumes clusters that are

(A) roughly spherical and similar in size

(B) of arbitrary shape

(C) one-dimensional

(D) labelled

Q.15 [MCQ • 1 Mark • Easy]

The centroid of the 2-D points $(1,1), (3,1), (2,4)$ is

(A) $(2,2)$ (B) $(3,1)$ (C) $(1,1)$ (D) $(2,4)$

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10. PCA & Dimensionality Reduction

14 questions • 9 MCQ, 1 MSQ, 4 NAT • Suggested time: 30 minutes

Q.1 [MCQ • 1 Mark • Moderate]

PCA finds directions that

(A) minimize the variance of the projected data (B) maximize the class separation

(C) minimize the number of samples (D) maximize the variance of the projected data

Q.2 [MCQ • 1 Mark • Moderate]

The principal components are the ___ of the data covariance matrix.

(A) rows (B) singular values (C) eigenvectors (D) diagonal entries

Q.3 [MCQ • 1 Mark • Moderate]

The amount of variance captured by each principal component is given by the corresponding

(A) data mean (B) eigenvalue of the covariance matrix

(C) eigenvector norm (D) number of features

Q.4 [NAT • 2 Marks • Moderate]

A covariance matrix has eigenvalues $6, 3, 1$. The fraction of total variance explained by the first principal component (two decimals) is _____.

Q.5 [NAT • 2 Marks • Moderate]

With eigenvalues $6, 3, 1$, the fraction of variance explained by the first *two* principal components (two decimals) is _____.

Q.6 [MCQ • 1 Mark • Easy]

Before applying PCA it is standard practice to

(A) remove the covariance

(B) center the data (subtract the mean), and often scale the features

(C) shuffle the labels

(D) add random noise

Q.7 [MCQ • 1 Mark • Moderate]

The principal components are mutually

(A) correlated (B) orthogonal (C) parallel (D) identical

Q.8 [MCQ • 1 Mark • Easy]

PCA is an ___ technique because it uses no class labels.

(A) reinforcement (B) semi-supervised (C) supervised (D) unsupervised

Q.9 [MCQ • 2 Marks • Hard]

For a classification problem, PCA has been used and it is proposed to use only the last (smallest-eigenvalue) principal components. This is generally a ___ idea. *(GATE DA 2026, Q.11 pattern.)*

(A) poor — those components carry the least variance / information

(B) neutral — variance is irrelevant

(C) good — they carry the most information

(D) required — PCA always uses the last components

Q.10 [MSQ • 2 Marks • Moderate]

Which of the following are true about PCA?

(A) The components are ordered by decreasing variance

(B) It can be computed via the SVD of the centered data matrix

(C) It is a linear dimensionality-reduction method

(D) It maximizes class separability

Q.11 [MCQ • 2 Marks • Hard]

Singular values σ_i of the centered data matrix relate to PCA eigenvalues λ_i (of the covariance, using divisor n) by

(A) $\lambda_i = \sqrt{\sigma_i}$

(B) $\lambda_i = n\sigma_i$

(C) $\lambda_i = \sigma_i$

(D) $\lambda_i = \sigma_i^2/n$

Q.12 [MCQ • 1 Mark • Moderate]

Whereas PCA is unsupervised, ___ is a supervised linear projection that maximizes between-class over within-class scatter.

(A) LDA (Linear Discriminant Analysis)

(B) the RBF kernel

(C) k-means

(D) logistic regression

Q.13 [NAT • 1 Mark • Easy]

Projecting d-dimensional data onto its top k principal components produces feature vectors of dimension _____.

Q.14 [NAT • 2 Marks • Moderate]

A total variance of 10 is spread over eigenvalues $\lambda_1 = 7, \lambda_2 = 2, \lambda_3 = 1$. To retain at least 85% of the variance, the minimum number of principal components needed is _____.

Part B — Previous-year questions (with official keys)

These are **genuine questions from the official GATE DA papers** in your archive. Each carries the **official GATE answer key** and its options are kept in the original paper order (not shuffled). Every question is tagged with its exact year and number. Grouped under the same topics as Part A.

2. Ridge Regression, Regularization & Bias–Variance

Q.1 PYQ [MCQ • 2 Marks • Moderate]

Which of the following statements is true for Ridge Regression?

(A) The regularizer in the objective function is used to guard against scenarios where the model works well for the test data but poorly for the training data.

(B) The regularizer of Ridge Regression uses the L_1 norm.

(C) Ridge Regression aims to reduce the number of parameters that have negative values.

(D) The regularizer may increase the bias of the model, but it helps in reducing the variance in predictions.

Source: GATE DA 2026, Q.37 (2 marks) — official paper & official answer key

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Q.2 PYQ [NAT • 2 Marks • Hard]

Linear Ridge Regression learns $y_{pred} = w^{\mathsf{T}}x$ with $w, x \in \mathbb{R}^2$, using Mean Absolute Error and a regularizer weight of 0.20 (on $\|w\|_2^2$). At an intermediate step $w = [-3.00, 4.00]^{\mathsf{T}}$. For input $x = [1.00, 2.00]^{\mathsf{T}}$ with true value $y_{true} = x_1 + x_2$, the overall regularized loss for this instance (rounded to two decimals) is _____.

Source: GATE DA 2026, Q.55 (2 marks) — official paper (accepted range 6.90–7.10)

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3. Logistic Regression & Classification Basics

Q.1 PYQ [MCQ • 1 Mark • Moderate]

Match the items in Column 1 with Column 2. Column 1: (p) Principal Component Analysis, (q) Naive Bayes Classification, (r) Logistic Regression. Column 2: (i) Discriminative Model, (ii) Dimensionality Reduction, (iii) Generative Model.

(A) (p)–(iii), (q)–(i), (r)–(ii)

(B) (p)–(ii), (q)–(i), (r)–(iii)

(C) (p)–(ii), (q)–(iii), (r)–(i)

(D) (p)–(iii), (q)–(ii), (r)–(i)

Source: GATE DA 2024, Q.18 (1 mark) — official paper & official answer key

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7. Neural Networks, MLP & Gradient Descent

Q.1 PYQ [NAT • 2 Marks • Moderate]

A fully-connected feed-forward multi-layer perceptron has 30 neurons in the input layer, followed by two hidden layers and an output layer. The first hidden layer has 4 neurons, the second hidden layer has 3 neurons, and the output layer has 1 neuron. No bias parameters are used. The number of

learnable parameters is _____.

Source: GATE DA 2026, Q.56 (2 marks) — official paper & official answer key

8. Model Evaluation & Cross-Validation

Q.1 PYQ [MSQ • 2 Marks • Hard]

20 stories of Author X and 10 stories of Author Y are mixed together and a classifier predicts the author of each. Out of X's stories, 6 were classified as Y; out of Y's stories, 2 were classified as X. Treating X and Y as two classes, which statement(s) is/are true?

(A) Accuracy of the classifier is 11/15.

(B) Precision of Class X is higher than the Precision of Class Y.

(C) Recall of Class X is higher than the Recall of Class Y.

(D) Accuracy of the classifier is 14/15.

Source: GATE DA 2026, Q.47 (2 marks) — official paper & official answer key

9. Clustering — k-means & Hierarchical

Q.1 PYQ [MCQ • 2 Marks • Moderate]

Four points in 3-D space are $P_1 = [2, 3, -1]$, $P_2 = [3, 1, 1]$, $P_3 = [5, -2, 3]$, $P_4 = [3, 3, 3]$. Hierarchical Agglomerative Clustering with *Manhattan* distance is used. Which two points are merged first?

(A) P_1, P_2 (B) P_2, P_3 (C) P_3, P_4 (D) P_2, P_4

Source: GATE DA 2026, Q.36 (2 marks) — official paper & official answer key

Q.2 PYQ [MCQ • 2 Marks • Hard]

Euclidean-distance k-means with $k = 3$ is run on 100 points. If the points $[1, 1]^\mathsf{T}$ and $[-1, 1]^\mathsf{T}$ are both in cluster 3, which one of the following points is *necessarily* also in cluster 3?

(A) $[0, 0]^\mathsf{T}$ (B) $[0, 2]^\mathsf{T}$ (C) $[2, 0]^\mathsf{T}$ (D) $[0, 1]^\mathsf{T}$

Source: GATE DA 2024, Q.19 (1 mark) — official paper & official answer key

10. PCA & Dimensionality Reduction

Q.1 PYQ [MCQ • 1 Mark • Moderate]

PCA is used to reduce dimensionality from 100 to 10 for a classification problem. What is the angle θ between the first and the tenth principal components?

(A) $\theta = 0°$

(B) $\theta = 90°$

(C) $90° < \theta \leq 180°$

(D) $0 < \theta < 90°$

Source: GATE DA 2026, Q.11 (1 mark) — official paper & official answer key

Answer Key

Answers only. A key with several letters (e.g. **AB**) is an MSQ, where every listed option must be selected and no other. For NAT questions any value within the usual GATE tolerance of the figure shown is accepted.

Part A — Practice

1. Linear & Multiple Regression

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	**2**	0.60	**3**	2.20	**4**	B	**5**	B
6	0.81	**7**	ABC	**8**	C	**9**	B	**10**	B
11	C	**12**	D	**13**	8	**14**	0.75	**15**	B
16	6	**17**	ACD	**18**	A	**19**	0	**20**	2
21	B	**22**	A						

2. Ridge Regression, Regularization & Bias–Variance

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	**2**	A	**3**	D	**4**	D	**5**	D
6	B	**7**	C	**8**	B	**9**	B	**10**	BCD
11	A	**12**	A	**13**	A	**14**	D	**15**	A
16	D	**17**	A	**18**	B				

3. Logistic Regression & Classification Basics

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	**2**	0.5	**3**	B	**4**	C	**5**	A
6	C	**7**	1.39	**8**	ABC	**9**	C	**10**	D
11	C	**12**	0.75	**13**	C	**14**	B	**15**	C
16	A								

4. Bayes & Naive Bayes Classifier

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	**2**	0.15	**3**	A	**4**	A	**5**	0.47
6	A	**7**	0.15	**8**	B	**9**	ACD	**10**	0.77
11	0.5	**12**	B	**13**	0.30	**14**	D		

5. Discriminant Analysis, k-NN & Decision Trees

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	1	**2**	0.5	**3**	0.32	**4**	0.72	**5**	B
6	0.60	**7**	A	**8**	B	**9**	B	**10**	5
11	7	**12**	B	**13**	A	**14**	D	**15**	A
16	BCD	**17**	D	**18**	0	**19**	1.58		

6. Support Vector Machines & Kernels

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	**2**	C	**3**	C	**4**	A	**5**	D
6	B	**7**	ABC	**8**	2	**9**	9	**10**	1
11	B	**12**	A	**13**	A	**14**	C	**15**	0

7. Neural Networks, MLP & Gradient Descent

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	35	**2**	43	**3**	D	**4**	D	**5**	B
6	B	**7**	3.2	**8**	D	**9**	ACD	**10**	B
11	D	**12**	C	**13**	4	**14**	A	**15**	C

8. Model Evaluation & Cross-Validation

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	0.70	**2**	0.80	**3**	0.67	**4**	A	**5**	0.73
6	A	**7**	5	**8**	B	**9**	B	**10**	D
11	ABD	**12**	A	**13**	A	**14**	D	**15**	0.25

9. Clustering — k-means & Hierarchical

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	**2**	A	**3**	4	**4**	8	**5**	D
6	A	**7**	C	**8**	A	**9**	A	**10**	ACD
11	B	**12**	C	**13**	C	**14**	A	**15**	A

10. PCA & Dimensionality Reduction

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	**2**	C	**3**	B	**4**	0.60	**5**	0.90
6	B	**7**	B	**8**	D	**9**	A	**10**	ABC
11	D	**12**	A	**13**	k	**14**	2		

Part B — Previous-year questions (official keys)

Source	Q	Official answer
GATE DA 2026, Q.37 (2 marks)	M2.1	**D**
GATE DA 2026, Q.55 (2 marks)	M2.2	**7.00**
GATE DA 2024, Q.18 (1 mark)	M3.1	**C**
GATE DA 2026, Q.56 (2 marks)	M7.1	**135**
GATE DA 2026, Q.47 (2 marks)	M8.1	**AB**
GATE DA 2026, Q.36 (2 marks)	M9.1	**D**
GATE DA 2024, Q.19 (1 mark)	M9.2	**D**
GATE DA 2026, Q.11 (1 mark)	M10.1	**B**

Score sheet

Anything below 70% on a topic means the concept, not the practice, is the problem — go back and revise before re-attempting.

Topic	Questions	Correct	Time taken
Linear & Multiple Regression	22		
Ridge Regression, Regularization & Bias–Variance	18		
Logistic Regression & Classification Basics	16		
Bayes & Naive Bayes Classifier	14		
Discriminant Analysis, k-NN & Decision Trees	19		
Support Vector Machines & Kernels	15		
Neural Networks, MLP & Gradient Descent	15		
Model Evaluation & Cross-Validation	15		
Clustering — k-means & Hierarchical	15		
PCA & Dimensionality Reduction	14		
Total	**163**		